

PRODUCT REQUIREMENTS DOCUMENT

Maple

Your campus, finally figured out.

A smart navigation app built for college students who keep getting lost between blocks, missing classes by three minutes, and asking strangers where the photocopy shop is.

Field	Detail
Document version	v1.0 — May 2026
Author	Priya, Product Manager
Status	Draft for review (MVP planning)
Target launch	MVP in 4 months, pilot at 1 campus, then 6–8 more in year one
Reviewers	Founding team, design lead, eng lead, faculty advisor
Built with	Notion, Figma, Claude, ChatGPT, and 31 student interviews

Internal draft. Please do not circulate outside the founding team.

Contents

Contents	2
1. The story so far	4
What this product is, in one paragraph.....	4
2. The problem we're solving	5
Things students told us, almost word-for-word.....	5
What we found in research	5
3. Who we built this for.....	7
3.1 Aisha — the new fresher	7
3.2 Rohan — the senior with back-to-back classes.....	7
3.3 Rhea — the student who uses a wheelchair	7
3.4 Mrs. Mehta — the visiting parent (secondary)	8
3.5 Karan — the events team lead (secondary)	8
4. What the research told us	10
Methods at a glance	10
Patterns that mattered.....	10
Things we explicitly chose not to build because of research	10
5. Vision, mission, and what we mean by winning.....	12
Vision (3 years out).....	12
Mission	12
How we'll know it's working (top-level)	12
6. Success metrics.....	13
North-star metric.....	13
Supporting product metrics.....	13
Counter-metrics (guardrails)	14
7. What Maple does	15
7.1 The map and search	15
7.2 The class schedule	15
7.3 The AI campus assistant	16
7.4 Accessibility	16
7.5 Campus life (the long tail).....	17
8. MVP scope	19
Hard 'yes' for MVP (must ship).....	19
Hard 'no' for MVP (post-pilot).....	19
9. Key screens (wireframes in Figma).....	20
Screen 1 — Onboarding (3 steps).....	20

Screen 2 — Home	20
Screen 3 — Navigation	20
Screen 4 — Search and the AI assistant	20
Screen 5 — Guest mode	20
Visual language.....	20
10. How it's built.....	22
11. How it makes money	23
12. Go-to-market.....	24
Year-one play.....	24
Year-two play.....	24
Why we'll win the pilot.....	24
13. Roadmap.....	25
14. What could go wrong	26
15. Open questions.....	27
16. Team and what we need	28
Budget	28
17. Appendix.....	29
A. Glossary	29
B. Research artefacts.....	29
C. Change log	29

1. The story so far

Most students don't realise how much time they lose to their own campus until they start tracking it. I asked thirty-one of them, across four colleges, to keep a one-week log of every time they got lost, asked for directions, or showed up late to a class because the lecture hall had moved. The average was 47 minutes a week. The highest, a first-year at a campus the size of a small town, was just over two hours.

Maple started there. The idea isn't to build another Google Maps. Outdoor walking directions across a 200-acre campus are already a solved problem, and students don't really need them. What they need is the messy stuff in between: which building has the print shop that's actually open, which staircase in the science block takes you to the third floor (the south one is bolted shut after 6 pm), where the accessible ramp is when the lift is down, and where their next class actually moved to because someone changed the timetable on Friday afternoon.

This document lays out what we're building, who it's for, what's in the MVP, what's not, and how we'll know it's working.

What this product is, in one paragraph

Maple in plain English

Maple is a phone-first app that helps a college student find anything on campus, get to it on time, and not look lost while doing it. It knows where their classes are, where their friends are (if they've shared), which rooms are crowded, and how to reach a place if they can't take stairs. It works indoors. It works without data on the campus Wi-Fi. And it gets smarter the more the campus uses it.

2. The problem we're solving

Campus life is logistical chaos. A typical day for a second-year undergrad in a mid-sized Indian university involves around eight to twelve location decisions: classrooms, labs, the canteen, a meeting with a TA in a faculty office no one can find, club practice across the field, a printout for an assignment. Each decision has friction. Each frustration is small. They add up.

And the existing tools don't help. Google Maps doesn't know your campus has buildings called “C-Block” and “New Block” right next to each other. The university PDF map is a relic. The college app, if there is one, was built by a final-year project team five years ago and hasn't been touched since.

Things students told us, almost word-for-word

“I've been here three years and I still don't know where Room 408 in the science block actually is. I just follow whoever else looks like a fourth-year.”

— Karthik, 4th year, Engineering

“My mom came for parent's day. She called me eleven times. Eleven.”

— Aisha, 2nd year, Liberal Arts

“There's a ramp to the library, but the one to the auditorium is on the back side and nobody tells you. I always end up going the long way.”

— Rhea, 1st year, uses a wheelchair

What we found in research

We ran 31 interviews and a survey of 412 students across four campuses. Some headline findings:

- **First-years burn the most time.** They spend roughly 3x as long getting to a destination in the first eight weeks compared to seniors. By month three, they've stopped using maps at all and mostly rely on memory and asking friends.
- **Indoor wayfinding is the real gap.** 78% of “I got lost” incidents happened inside a building, not finding the building.
- **Class schedule changes are silent killers.** Almost everyone has a story about showing up to the wrong room because a lecture got moved and the email got buried.
- **Accessibility is invisible.** Students with mobility needs we spoke to all had a mental map of “which routes work” that they'd built themselves over months. None of it is documented anywhere.
- **Parents and visitors don't exist in current tools.** There is no campus where a guest can walk in and orient themselves without asking a student. This was the single most repeated frustration during parent's day and admissions visits.

Why this is worth building

Campus navigation isn't a 'nice to have'. It's the first product on a student's phone every morning and the last one at night. Whoever owns that surface owns a lot more than wayfinding — events, lost-and-found, club discovery, food on campus, study group formation. We're not just solving for getting from A to B. We're solving for the daily operating system of college life, with navigation as the wedge.

3. Who we built this for

Five personas. Three primary, who use Maple every day. Two secondary, who use it occasionally but matter a lot for the business side.

3.1 Aisha — the new fresher

Aisha, 18, first year	Just joined a 250-acre campus three weeks ago. Comes from a small town. Smartphone-first.
What they're trying to do	Get to the right class on time, find food, not look like a fresher.
What's annoying right now	Confuses buildings with similar names. Asks seniors for directions and feels embarrassed. The college app is broken on her phone.
What success looks like for them	She opens Maple, sees her class is in C-204, and the app says “Five minutes from where you are. Take the back stairs in B-block, it's faster.”

3.2 Rohan — the senior with back-to-back classes

Rohan, 21, fourth year	Knows the campus cold. Has a packed schedule with classes across far-apart blocks.
What they're trying to do	Make it from a lab in one building to a lecture in another in nine minutes.
What's annoying right now	Doesn't need a map; he needs a heads-up when a class moves. Right now he finds out by walking into an empty room.
What success looks like for them	Maple pings him: “Your 11 am is in NB-301 today, not the regular room. Leave by 10:53 to make it.”

3.3 Rhea — the student who uses a wheelchair

Rhea, 19, second year	Has built her own mental map of accessible routes after months of trial and error.
What they're trying to do	Get anywhere on campus without dead-ending at a flight of stairs.
What's annoying right now	Lifts are unreliable. Ramps exist but aren't marked. She has to plan trips around guesswork.
What success looks like for them	Maple offers her an accessible-only route by default. When a lift breaks, the app reroutes within seconds, and the maintenance team gets pinged too.

3.4 Mrs. Mehta — the visiting parent (secondary)

Mrs. Mehta, 48, parent	Visits campus twice a year. Has never used the college app.
What they're trying to do	Find her daughter, the auditorium, the parking, and the bathroom — ideally without bothering her daughter.
What's annoying right now	Signage is inconsistent. Security guards are friendly but vague. She ends up calling her daughter every twenty minutes.
What success looks like for them	She scans a QR at the main gate. Maple opens a stripped-down “guest mode” and walks her to the auditorium in three taps.

3.5 Karan — the events team lead (secondary)

Karan, 22, placement & events committee	Runs five major events a year. Coordinates rooms, signage, food stalls.
What they're trying to do	Make sure 500 attendees end up at the right place, on time, without his phone melting from WhatsApp messages.
What's annoying right now	Spends event days posting paper signs and replying “Ground floor, behind the canteen” to forty people an hour.

Karan, 22, placement & events committee	Runs five major events a year. Coordinates rooms, signage, food stalls.
What success looks like for them	He sets up the event in Maple's organiser console. Attendees get auto-routing. He gets a live attendance and flow heatmap instead of a phone full of texts.

4. What the research told us

Before any feature decisions, we ran a four-week discovery phase. Mix of methods: interviews, surveys, a small diary study, and time-on-campus shadowing. The point wasn't to confirm what we already thought. It was to find the moments where students gave up, asked for help, or built workarounds.

Methods at a glance

Method	What we did
Interviews	31 in-person, 45 minutes each. Mixed years, mixed majors, mixed campuses.
Survey	412 responses across four campuses, distributed via campus reps and WhatsApp.
Diary study	12 students kept a one-week navigation log. We paid them in chai vouchers.
Shadowing	5 half-days walking with students through their normal day.
Stakeholder chats	Conversations with two facilities heads, three faculty members, and one accessibility office.

Patterns that mattered

A few findings cut across every campus we visited.

- **The pain is daily, not occasional.** Students don't have a navigation problem once a semester. They have ten small ones every day. That changes how we design.
- **People matter more than maps.** Almost every interviewee said they get directions by asking a person, not opening a map. Any product that ignores this is fighting the user.
- **The schedule is the killer feature.** When we showed wireframes, the moment students lit up was always the same: “it knows where my next class is and tells me when to leave.”
- **Trust is local.** Students believe campus info that comes from other students. They are quietly skeptical of admin-published data, which is often outdated.
- **Privacy is a hard line.** Location sharing with friends, even opt-in, makes them nervous. “Not tracking, just nearby” was a phrase that came up a lot.

Things we explicitly chose not to build because of research

Discovery is also about killing ideas. We dropped the following before they hit the spec:

-
- A full-screen AR turn-by-turn overlay. Students said they'd never hold up their phone for that long. Battery, awkwardness, and headphones-everywhere reality made it a no-go.
 - A campus social network. Every student we asked already has Instagram and WhatsApp. Nobody wanted a sixth feed.
 - A mandatory check-in system for classes. Faculty wanted it; students hated it. Not our hill to die on.

5. Vision, mission, and what we mean by winning

Vision (3 years out)

Maple is the daily-driver app on every college student's home screen. It's how they navigate, plan their day, and stay in the loop on what's happening around them. If you go to college, you have Maple.

Mission

Make every square foot of campus knowable, accessible, and easy to reach — starting with navigation, expanding to the rest of campus life.

How we'll know it's working (top-level)

- A first-year student installs Maple in their first week, opens it daily by week three, and never uninstalls it.
- Students at our partner campuses save measurable time — we're targeting a 30%+ drop in “got lost” incidents and class lateness.
- Universities renew their license year-on-year because students complain when it's gone.

6. Success metrics

We track three layers of metrics: a north-star, supporting product metrics, and counter-metrics that act as guardrails.

North-star metric

Weekly Class Routings per Active Student (WCR)

Every time a student is routed to a class, lab, or campus location, that's one routing. The metric captures actual daily utility, not vanity sign-ups.

Target: 12 routings per active student per week by month 6 of pilot. (For context: that's around 2–3 per weekday, which matches a typical student timetable.)

Supporting product metrics

Metric	Target (12 months in)	Why this one
DAU / MAU ratio	≥ 0.55	Stickiness. Tells us if students open it daily, not just on event days.
Task completion rate (got to destination)	≥ 92%	If they don't make it where they're going, the product is broken.
Median time-to-destination vs. baseline	30% faster	Measured against pre-Maple self-reported times.
On-time class arrival rate	≥ 95%	Directly tied to the schedule routing feature.
30-day retention (new users)	≥ 70%	The fresher acquisition story.
Accessible-route usage	100% of users who opt in route via accessible paths	Non-negotiable. If it fails for them, the product fails.
First-week activation (set schedule + first route)	≥ 80%	If they don't activate in the first week, they probably won't.

Metric	Target (12 months in)	Why this one
University NPS (admin buyers)	≥ 50	Renewals depend on this.

Counter-metrics (guardrails)

- False routing rate (sent to wrong place) under 0.5%. Above that and we have a trust problem we can't fix with features.
- Battery drain in normal use under 5% per hour. Anything more and students will uninstall.
- Push notifications opt-out rate under 15%. If we annoy people, this is where we'll see it first.
- Privacy complaints: zero. Genuinely zero. One incident sets the product back six months.

7. What Maple does

The product breaks into five surfaces. Each maps to a real moment in a student's day.

7.1 The map and search

This is the spine of the product. Without it, nothing else matters.

ID	Feature	What it does	Who it's for	MVP?
F-1.1	Outdoor map	Branded, lightweight campus map with all buildings, paths, parking, gates	Everyone	Yes
F-1.2	Indoor maps	Floor-by-floor maps for major buildings. Search by room number.	Everyone	Yes (top 6 buildings)
F-1.3	Smart search	“Where can I print?”, “Nearest washroom”, “Ms. Sharma's office” — typed or voice	Aisha, Mrs. Mehta	Yes (typed only)
F-1.4	Walking route with ETA	Step-by-step, with realistic time estimates calibrated to actual student walking speed	Everyone	Yes
F-1.5	Offline mode	Full map and routing work on campus Wi-Fi or no data at all	Everyone	Yes

7.2 The class schedule

The reason most students will open the app every day.

ID	Feature	What it does	Who it's for	MVP?
F-2.1	Auto-import timetable	Pulls a student's timetable from the university's system or a one-time CSV upload	All students	Yes
F-2.2	“Leave by” reminders	Push notification that tells you when to start walking to make it on time	Rohan, Aisha	Yes

ID	Feature	What it does	Who it's for	MVP?
F-2.3	Room change alerts	If a class moves, the app updates and notifies. Pulled from faculty announcements or admin updates.	Everyone	Yes
F-2.4	Class day view	A clean view of today, with each class, room, route preview, and time-to-leave	Everyone	Yes

7.3 The AI campus assistant

A chat-style helper that answers the questions students currently ask each other in WhatsApp groups.

ID	Feature	What it does	Who it's for	MVP?
F-3.1	Free-text Q&A	“Is the library open after 10?” “Who's the HOD for ECE?” “Where can I get my ID card reissued?”	Everyone	Yes (limited corpus)
F-3.2	Grounded answers	Every answer cites the source — office, page, person — so students can trust it	Everyone	Yes
F-3.3	“Ask a senior” fallback	If the AI doesn't know, students can opt to ask the senior volunteer pool. Optional.	Aisha	Phase 2
F-3.4	Voice mode	Hands-free, walk-and-talk	Everyone, especially Rhea	Phase 2

7.4 Accessibility

This is the part of the product I care most about getting right.

ID	Feature	What it does	Who it's for	MVP?
F-4.1	Accessible routing	Routes that avoid stairs, prioritise lifts, ramps, and step-free paths	Rhea, anyone temporarily injured	Yes
F-4.2	Live lift / ramp status	Crowdsourced + facilities-fed status of every lift and ramp	Everyone who needs it	Yes
F-4.3	Visual accessibility	Large-type mode, high-contrast theme, screen-reader-clean	Low-vision students	Yes
F-4.4	Sensory-friendly routes	Quieter paths that avoid loud zones (canteens at peak hours, generator rooms)	Students who need them	Phase 2

7.5 Campus life (the long tail)

Once we've earned the home-screen slot, we add the surrounding layer that turns Maple into a campus companion.

ID	Feature	What it does	Who it's for	MVP?
F-5.1	Crowd & queue predictions	How crowded is the canteen right now? Library at 8 pm?	Everyone	Phase 2
F-5.2	Events & clubs	Discover events nearby, RSVP, get auto-routing the day-of	Everyone, especially first-years	Phase 2
F-5.3	Lost & found	Photo + location-based posts, with notifications to people who passed through that zone	Everyone	Phase 2

ID	Feature	What it does	Who it's for	MVP?
F-5.4	Friends nearby (opt-in)	See which friends are on campus right now. No live tracking, just presence.	Most students	Phase 3
F-5.5	Guest mode	QR-launched stripped-down map for parents and visitors. No login.	Mrs. Mehta	Yes
F-5.6	Organiser console	For event organisers and admin: set up events, see flow heatmaps, send routing nudges	Karan, events team	Phase 2

8. MVP scope

We're not shipping all of section 7 at once. The MVP exists to answer one question: do students at one campus open this app every day and tell their friends about it? If yes, we expand. If no, we know what to fix.

Hard 'yes' for MVP (must ship)

- Outdoor map for the pilot campus, plus indoor maps for the six highest-traffic buildings.
- Smart typed search across rooms, offices, services, common queries.
- Class schedule import, day view, and "leave by" notifications.
- Room change alerts (admin- or faculty-pushed).
- Walking routes with ETA, including accessible routing with lift/ramp status.
- AI campus assistant with a curated knowledge base (~300 FAQs at launch).
- Guest mode via QR at the main gate.
- Offline mode on the pilot campus Wi-Fi.
- Basic onboarding: phone number sign-in, pick your year, paste timetable.

Hard 'no' for MVP (post-pilot)

- Crowd predictions and live heatmaps.
- Events, clubs, lost & found.
- Friends-nearby (privacy work is heavy; this is a v2 feature done properly or not at all).
- Voice mode for the AI assistant.
- Organiser console.
- Multi-campus support (each campus needs its own map data; we do this one at a time).

The bet

If a first-year student doesn't open Maple at least three times in their first week, the MVP failed even if every feature works. The whole MVP is built around earning that habit.

9. Key screens (wireframes in Figma)

Full Figma file lives at: [figma.com/file/\[link\]](https://figma.com/file/[link]). Below is a quick walkthrough so a reader doesn't need to open Figma to understand the flow.

Screen 1 — Onboarding (3 steps)

- Step 1: phone number or college email. OTP. No passwords.
- Step 2: pick campus, year, branch. Pre-filled if SSO is connected.
- Step 3: import timetable. Three options: SSO auto-pull, photo of timetable, or paste.

Screen 2 — Home

The home screen is the day's calendar layered with location. At the top: next class, room number, time-to-leave countdown, a single big button that says “Get me there.” Below: the day in a vertical timeline. Tap any class to preview the route. There's a small chat bubble bottom-right that opens the AI assistant. The map sits as a peek-through layer behind everything, so it always feels grounded in place.

Screen 3 — Navigation

Walking-mode is split into two halves. Top half: the map. Bottom half: the next instruction in large type, ETA, and a fallback “Not sure where I am” button that reveals nearby landmarks for self-orientation. Accessible-route users see a small pink badge that says “Step-free route.”

Screen 4 — Search and the AI assistant

One unified search bar. Typing pulls up structured results (rooms, offices, services). Hitting a small sparkle icon switches to the AI assistant, where the same query becomes natural language. Every answer ends with a “Take me there” button if the answer is a place.

Screen 5 — Guest mode

Scanning a QR at the gate opens a stripped-down version. No login, no account. Three big tiles: “Find a building,” “Find a person,” “Parking and washrooms.” Done.

Visual language

Soft pinks, generous whitespace, friendly geometric icons, rounded corners. The map itself uses a warm beige base, pink for primary routes, deeper rose for accessible routes. Type is set in a humanist sans (Inter at launch). The aim is to feel like a campus app a student would actually want on their home screen, not a utility.

10. How it's built

A short technical sketch. Detailed engineering design lives in a separate doc.

Layer	Approach
Mobile app	React Native (single codebase for iOS + Android). Native modules for Bluetooth beacon scanning.
Indoor positioning	Hybrid of Wi-Fi RSSI fingerprinting and Bluetooth beacons in the six MVP buildings. GPS outdoors.
Map data	Custom GIS pipeline. Each campus gets a guided mapping pass with the facilities team. Stored as MapLibre vector tiles.
Routing engine	Open-source pgRouting on top of PostGIS, with custom edge weights for accessibility and indoor segments.
AI assistant	Retrieval-augmented generation over a curated campus knowledge base. We're model-flexible — Claude or GPT depending on cost and quality — with grounding required on every answer.
Schedule integration	Admin connector to the university's LMS / ERP (SAP, Moodle, etc). Fallback: photo OCR or paste.
Backend	Python (FastAPI), Postgres, Redis. Hosted on AWS, single region per cluster of campuses for data residency.
Real-time	Push notifications via FCM and APNs. Lightweight WebSocket layer for live status changes (lift down, room moved).
Privacy	Location data stays on-device by default. Anything that leaves the device is aggregated and stripped of identifiers.

11. How it makes money

Maple is free for students. Always. The product fails the moment a student is asked to pay to find their classroom. Revenue comes from three other places.

Revenue stream	Detail
University license	Annual fee per campus, priced by enrolled student count. Includes setup, mapping, integration with their systems, admin console, support. Ballpark: ₹15–40 lakh / year per campus depending on size.
Sponsored campus listings	Cafes, print shops, bookstores on or near campus pay to be featured in search results and on the map. Disclosed clearly. Light-touch.
Premium for visitors / alumni	Optional Maple+ tier for parents during admissions / parents' day, and alumni for reunions. ₹199 one-time. Doesn't change what students get.

Why this structure

Selling to universities solves the cold-start problem: they fund the mapping work and act as the distribution channel for student adoption. Sponsored listings are an upside, not a foundation. Anything that asks students for money would destroy trust.

12. Go-to-market

Year-one play

- Pilot at one large campus where we have a faculty advocate. Free for year one in exchange for being a design partner.
- Heavy on-the-ground presence during orientation week. We want first-years to install Maple before they get their ID cards.
- Build a senior-volunteer program: 30–50 senior students per campus who help curate the knowledge base. Light incentive (resume mention, free merch).
- Aim for organic word-of-mouth, not paid acquisition. Campus products spread person-to-person or not at all.

Year-two play

- Expand to 6–8 paying campuses, prioritising those that already pinged us during pilot demos.
- Launch the alumni and parents tier.
- Start building the organiser console for events teams.

Why we'll win the pilot

- We chose the campus carefully. It's a tier-1 college with a strong digital culture and a facilities team that's already frustrated with the status quo.
- The competitive set is weak. Existing college apps are abandoned. Google Maps doesn't cover indoor.
- Accessibility is a real differentiator and a strong reason for the administration to back us publicly.

13. Roadmap

Phase	When	What ships
Discovery	Months 0–1	Interviews, surveys, faculty conversations, MVP scope locked
Design & spec	Months 1–2	Figma flows, eng design doc, faculty sign-off
MVP build	Months 2–5	Build everything in section 8. Mapping pass at the pilot campus.
Closed beta	Month 5	200 students, two weeks, fix what breaks
Pilot launch	Month 6 (timed to start of new semester)	Full pilot campus rollout
Iterate	Months 6–9	Weekly releases driven by usage data and student feedback
Phase 2 features	Months 9–12	Crowd predictions, events, organiser console, voice mode
Second campus	Month 12	First paying campus live

14. What could go wrong

Every PRD has a risks section. Most of them are wishful and vague. Here's a more honest list.

Risk	How worried	What we do about it
The university timetable system is a mess we can't integrate with	Very	Build the photo + paste fallback first. Treat SSO as a bonus, not a requirement.
Indoor positioning is jittery in the first few weeks	High	Aggressive in-app messaging that sets expectations. Crowdsourced corrections. Only roll out to six buildings, not all.
Students don't see the point during pilot week	High	Orientation-week presence, free merch for installing, ride-along with student council.
Faculty resist because they think we're surveilling them	Medium	Be upfront. No teacher-tracking. Talk to faculty senate before launch.
A privacy incident, real or perceived	Catastrophic if it happens	On-device location by default, no friend tracking in MVP, third-party privacy review before pilot.
Lifts break and our 'lift down' status is wrong	Medium	Crowdsourced reporting + facilities feed. Show last-updated timestamp on every status.
Bigger players (Google, college ERPs) ship a competitor	Medium	Our wedge is depth and student-friendliness. Move fast on indoor + accessibility. Defend by becoming the daily habit.

15. Open questions

Things we haven't decided yet. Listed honestly so reviewers can push back.

#	Question	Owner	Need by
1	Do we launch the AI assistant in the MVP, or wait until the knowledge base is bigger?	PM + Design	End of design phase
2	Do we offer the app in regional languages from day one, or English-only for MVP?	PM	Before build starts
3	Should the senior-volunteer program be paid (small stipend) or recognition-only?	PM + GTM	Pre-pilot
4	Phone-number sign-in or college-email-only? Trade-off between friction and abuse prevention.	Eng + PM	Before build
5	How aggressive are we about indoor coverage at launch? Six buildings or all of them at lower quality?	PM + Eng	Mapping pass
6	What's our policy for ads / sponsored listings? Hard rules now, before we need the revenue.	PM + Founder	Pre-pilot

16. Team and what we need

Role	Headcount	Notes
Product (me)	1	Owns spec, research, success metrics
Design	2	1 product designer, 1 visual / brand designer (mostly for the map)
Mobile engineering	3	React Native + native modules
Backend engineering	3	APIs, routing engine, integrations
GIS / mapping	1	Owns the campus mapping pipeline
AI / ML	1	Builds the assistant, owns evaluation
QA	1	Becomes a need from month 3
Founder / GTM	1	Sells the pilot, owns university relationships
Total Y1	13	

Budget

- Year-one all-in: roughly ₹2.8–3.2 crore (people-loaded), ₹15–20 lakh infra and tooling, ₹10 lakh mapping hardware for first three campuses.
- Revenue in year one is small by design. We're optimising for the second-year story.

17. Appendix

A. Glossary

Term	Meaning
WCR	Weekly Class Routings per active student — our north-star metric
DAU / MAU	Daily / monthly active users. Their ratio is a stickiness measure.
Indoor positioning	Locating someone inside a building, where GPS doesn't reach. Done with Wi-Fi signal strength and Bluetooth beacons.
Accessible routing	Routes that avoid stairs and prefer step-free paths, lifts, and ramps.
Guest mode	A login-free, stripped-down version of the app for parents and visitors.
Pilot campus	The first university we launch on. Free for year one in exchange for being a design partner.

B. Research artefacts

- 31 interview transcripts (anonymised) — in Notion
- Survey results (412 responses) — raw and summarised
- Diary study logs from 12 students — in Notion
- Shadowing field notes — in Notion
- Faculty and facilities conversation notes — in Notion

C. Change log

Version	Date	Author	What changed
v0.1	Apr 2026	Priya	First draft, problem statement, personas
v0.4	Apr 2026	Priya	Added research synthesis, MVP scope debate
v0.8	May 2026	Priya	Reviewer pass; metrics tightened, risks rewritten

Version	Date	Author	What changed
v1.0	May 2026	Priya	Draft for review